Department of Computing Technology and Engineering

Assignment on Generative Models in Deep Learning

Focus: GANs, Autoencoders, and Variational Autoencoders

Learning Objectives

By completing this assignment, students should be able to:

- Understand **generative modeling** in deep learning
- Explain and differentiate **Autoencoders, Variational Autoencoders (VAEs), and GANs**
- Analyze **architectures, loss functions, and training challenges**
- Implement or experiment with at least one generative model
- Critically evaluate applications and limitations

◆ PART A: Conceptual Foundations (30 Marks)

Question 1: Generative Models Overview (10 Marks)

- Define **generative models** in deep learning
- Differentiate between:
  - Generative vs Discriminative models
- Explain **3 real-world applications**

Question 2: Autoencoders (10 Marks)

- Explain:
  - Encoder and Decoder structure
  - Latent space representation
- Types of Autoencoders:
  - Vanilla Autoencoder
  - Denoising Autoencoder
  - Sparse Autoencoder
- What problem do Autoencoders solve?

Question 3: Variational Autoencoders (VAEs) (10 Marks)

- Explain:
  - The key idea behind VAEs (probabilistic encoding)
  - Latent space as a distribution
- Define:
  - Reparameterization trick
  - KL Divergence
- Why are VAEs better than traditional Autoencoders for generation?

◆ PART B: Generative Adversarial Networks (30 Marks)

Question 4: GAN Architecture (10 Marks)

- Explain the two components:
  - Generator
  - Discriminator
- Describe how they interact (adversarial training)

Question 5: GAN Loss Function (10 Marks)

Use the GAN objective:

- Explain each term in simple language
- What is the goal of:
  - Generator?
  - Discriminator?

Question 6: Challenges in GAN Training (10 Marks)

Discuss:

- Mode collapse
- Training instability
- Vanishing gradients
- Evaluation difficulties

◆ PART C: Comparison and Critical Thinking (20 Marks)

Question 7: Critical Analysis (10 Marks)

- When would you choose:
  - Autoencoder instead of GAN?
  - VAE instead of GAN?

- Justify with examples

◆ PART D: Practical Component (20 Marks)

Implement **one** of the following using **TensorFlow or PyTorch**:

- Autoencoder on MNIST
- VAE on MNIST
- Simple GAN for image generation

Include:

- Model architecture
- Training process
- Output samples (images)
- Observations

Report Requirements (10 Marks)

- Length: **3–4 pages**
- Must include:
  - Diagrams of architectures
  - Clear brief (straight to the point) explanations
  - Figures (generated outputs if coding)
  - References (IEEE or APA style)

Bonus (Up to +5 Marks)

- Explore advanced GAN variants:
  - DCGAN
  - CycleGAN
  - StyleGAN
- OR discuss ethical concerns (deepfakes, misuse)